

The Future is Small

THE TERMS

A nanometre is a billionth of a metre—about five atoms wide. Nanotechnology brings together physics, chemistry, electronics, computer science and biotechnology. It refers to the technology that designs and fabricates nanosystems—systems on the nanoscale, that is, on a scale of nanometres. A nanocomputer is one that uses components only a few nanometres large.

HOW DO YOU BUILD A NANOCOMPUTER?

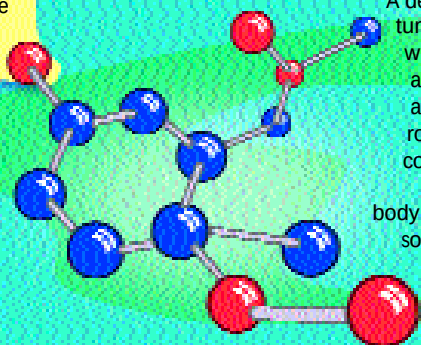
There are four different approaches to building them. The first is electronic: Keep shrinking transistors, and then do something different when they reach the nanoscale.



The second employs quantum physics:

A desktop quantum computer would be millions of times faster than your desktop, and might even operate in other universes. The third approach is mechanical—imagine tiny gears, tiny robotic arms and so forth, floating around, doing computations and other things.

The fourth kind is the weirdest: The cells in your body are probably doing some computing to stay alive, so someone thought of using cellular material to build a computer.



GRAY GOO

"Gray goo: n. A hypothetical substance composed of large numbers of sub-micron-sized, self-replicating robots programmed to make copies of themselves out of whatever is available. The image that goes with the term is that of the Earth's entire biosphere being eventually converted to robot goo. This is the simplest of the nanotechnology disaster scenarios ..." (from *'The Jargon Dictionary,'* <http://info.astrian.net/jargon/>)

After some scientists began looking seriously at the idea of building nanobots that would manipulate nanothings, Sun co-founder, Bill Joy, came up with this: "An immediate consequence of the ... bargain in obtaining the great power of nanotechnology is that we run a grave risk—the risk that we might destroy the biosphere on which all life depends." Joy also said that left unchecked, self-replicating molecular machines could endlessly copy themselves, engulfing the globe in a nanotechnological 'gray goo.'

Gray goo has been a worry to some people. Doomsday theories about nanobots are rife. But D-day apart, there are scientists who believe that nanobots can't ever be built. Some, such as Caltech physicist Stephen Quake, simply dismiss nanobots as science fiction. Others quote quantum mechanical principles to prove that we can never have control at the atomic level. However, the strongest argument against the possibility of nanobots comes from Richard Smalley, a Nobel prize-winning chemist. In 1999, he was a proponent of nanotechnology, but in 2001, he wrote in an article in *Scientific American* that such control would not be possible.

Inventions such as the DNA hand at LMU (see *'Recent Developments'*) will heat up the debate.

Remember, the DNA hand, according to its creators, will be capable of building nanothings a couple of years from now!

